

Yuan Huang 黄远

huangyuan4629@gmail.com

Homepage · Google Scholar · GitHub

Northeastern University, Shenyang, China

RESEARCH INTERESTS

Natural language processing, with a focus on **measurement validity** — whether the benchmarks and automatic judges used to rank models measure what they claim to. One result recurs across my projects: **surface competence $\neq$ underlying competence**. I want to measure that gap, explain it mechanistically, and close it. My current project is on the **interpretability of bias in multimodal models**: identifying the internal semantics that carry a bias, and mechanisms to correct it.

EDUCATION

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) — Visiting Student 2026 – 2027

– Advised by Xiuying Chen (Assistant Professor, MBZUAI)

Northeastern University, China — B.Eng., Computer Science and Technology 2023 – 2027 (expected)

– CGPA 4.0481 / 5.00 (90 / 100) · Rank 5 / 110 (top 4.55%) in major cohort

– Advised by Shiqi Zhao (Assistant Professor, Northeastern University) since 2024

PUBLICATIONS

[1] Do MLLM Judges Judge the Edit? Auditing Bias in Image Editing Evaluation with Verified Quality Preservation

Yuan Huang*, Zirui Song*, Xiuying Chen (** equal contribution*)

EditJudgeBias · Under review at ICLR 2027 · co-first author · arXiv:2610.01670, Oct 2026

[2] Two Routes to the Middle: Placement Search and Brain Readouts Converge on Where Continual Learners Should Specialize

Yuan Huang, Zihan Chen, Runbin Zhang, Hongwei Ding, Changzeng Fu, Shiqi Zhao

LS-B · Under review at ICLR 2027 · first author · arXiv:2610.01590, Oct 2026

[3] Beyond Survival: Evaluating LLMs in Social Deduction Games with Human-Aligned Strategies

Zirui Song*, Yuan Huang*, Junchang Liu*, et al. (** equal contribution*)

WereBench · EMNLP 2026, Main Conference · co-first author · arXiv:2510.11389, Oct 2025 · Dataset

[4] Geolocation with Real Human Gameplay Data: A Large-Scale Dataset and Human-Like Reasoning Framework

Zirui Song, Jingpu Yang, Yuan Huang, et al.

Geolocation · Preprint · arXiv:2502.13759, Feb 2025 · Dataset · Code

RESEARCH EXPERIENCE

EditJudgeBias — auditing bias in MLLM judges for image editing

Oct 2026

Co-first author · under review at ICLR 2027

- **Design.** 1,196 real editing samples and 13 cues across four evaluation sites. A shift counts as bias only if the cue is verified to preserve edit quality, and is read against the judge's own zero-dose and re-query noise floors, not against zero.
- **Findings.** Quality-preserving cues move all five MLLM judges beyond their own noise; swapping candidate order alone reverses up to 60.9% of pairwise decisions, against 1.0–10.5% on re-query.
- **Rigor.** 1,396 tests, 117 sha256-stamped artifacts; the audit overturned four of my own conclusions.

LS-B — brain readouts for layer specialization in continual learning

Oct 2026

First author · under review at ICLR 2027

- Placement search over ViT depth traces an inverted U (up to 3.5 pp); LS-B reads early tasks through an fMRI encoding model of twelve human visual areas and picks blocks overlapping the peak — on Split ImageNet-R, within 1.5 pp of full-BiLoRA at 60% of the storage, with no labels or backpropagation.

WereBench / WereAlign — human-aligned evaluation of LLM social reasoning

Oct 2025

Co-first author · inception through submission · EMNLP 2026 Main

- Benchmark built from 100+ hours of professional human social-deduction gameplay (32.4M utterance tokens, 30 roles, 15 rule variants), scored against winning-side MVP trajectories rather than LLM self-play; 5 verbal dimensions and 2 decision tasks.
- Half of the evaluated models score below 0.50 (best 0.720). Deception reasoning and counterfactual trade-off are weakest while persuasive generation stays strong — fluency ≠ strategic correctness.

Geolocation — GeoComp / GeoCoT / GeoEval, from real human gameplay data

Feb 2025

Co-author (3 of 9) · preprint

- Dataset from a public geo-guessing platform — 740K users, 2.7M locations, 25M human response records — with human accuracy calibrating item difficulty. Multi-step geographic reasoning (GeoCoT) improved classic geolocation metrics by up to 25%, and reasoning quality as scored by GeoEval by 9%.

HONORS & AWARDS

- China National Scholarship — national level · 2023–2024
- Northeastern University Comprehensive Scholarship · 2024–2026
- Third Prize, National Finals, Lanqiao Cup Programming Competition · 2024

SKILLS

- Python, C++, Java · PyTorch, Hugging Face Transformers · backend engineering intern, 2024 (Java / Spring Boot)